

Food Healthiness Classification Based on Ingredients

GitHub link: <https://github.com/Bubcheeseburger/DS4002-CS3>

Context:

In the modern world, there is increasing pressure to make “healthy” diet choices, but most consumers rely almost entirely on the Nutrition Facts label and the numeric values printed on packaging. This approach overlooks a key part of health assessment, that being the ingredients list. It is possible for two foods to have similar calorie or fat counts, but differ dramatically in quality depending on additives, preservatives, sweeteners, and processing agents. The effects of these hidden agents are not clearly represented by these labels, yet they can strongly influence metabolic response, inflammation, and long-term health outcomes. Machine learning provides an opportunity to assess food quality in a way that does not require a background in nutrition, making healthier decision-making more accessible by automatically gauging health from ingredient lists.

Motivation:

You are a data scientist working with a health-focused consumer advocacy group that aims to help individuals make informed food choices quickly and easily. They have observed that most shoppers do not have the expertise to interpret ingredient lists, and current “healthy” labels can be misleading or influenced by marketing. The organization wants to explore whether food products can be classified as healthy, intermediate, or unhealthy based solely on their ingredient compositions. If successful, this approach could encourage transparency in food labeling and empower consumers to evaluate products without needing to decode complex nutritional terminology.

Deliverable:

The student will develop a system that can determine whether a packaged food product is healthy, intermediate, or unhealthy based on its ingredient list. To do this, the student will first design a set of ingredient-based rules that assign relative health scores and then use those labels to train and evaluate a Random Forest Classifier. The final deliverable will include the classifier’s predictions and a performance summary that demonstrates the validity and accuracy of the model.